

F-5. Forest Silviculture Operations				
Forest / Shrub Land Conservation	Medium	High		Implementation Time of Measures
Measure Type	Cost	Preferences by Community	Hazards	Implementation Time of Measures

Descriptions									
Measures	The methods and strategies used to control the formation, development, composition, health, and regeneration of forests are referred to as forest silviculture operations. These activities are intended to accomplish particular goals, including lumber production, biodiversity conservation, or ecosystem restoration, while simultaneously preserving the health and productivity of forests. The science and practice of controlling forest ecosystems to achieve ecological and financial objectives is known as silviculture.								
	In order to enhance the health, productivity, and ecological value of the forests, the CHT region's forest silviculture operations prioritize sustainable forest management. Adopting particular silvicultural practices that guarantee long-term forest sustainability while meeting the socioeconomic needs of the area is crucial because of the region's high terrain, diverse soil types, and climate, which provide particular obstacles.								
Location	R-114	R-99	R-65	K-98	K-76	K-71	B-173	B-77	B-76
	■	■	■	■	■	■	■	■	■
Applicable Sites	Suitable for degraded lands, barren area, low forest								
Slope	Less than 3%		3% to 15%		15% to 30%		30% to 60%		Above 60%
	√		√		√				
Design options and performance	• Carefully removes mature trees while preserving younger trees and natural regeneration. • Gradual removal of overstory trees in stages to allow natural regeneration of desired species. • Harvesting trees in designated areas while maintaining riparian buffer zones and wildlife corridors. • Integrates forest trees with agricultural crops or livestock. • Supports the natural regrowth of forests by protecting existing seedlings and removing invasive species.								
Feasibility criteria	• Assessing soil fertility and implementing soil conservation techniques, such as contour plowing and terracing, are necessary to stop erosion and preserve soil health. Additionally, forest silviculture operations need to choose species that are resilient to adverse environmental circumstances, like intense monsoon-season rainfall, and that are appropriate to the local environment. • Particularly for high-value species like teak, species selected for silviculture operations must have high growth rates, insect resistance, and excellent								

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	timber quality. Plantations with mixed species can improve soil health, lower the danger of pests and diseases, and support the sustainability of forests. The purpose of forest silviculture should be to preserve biodiversity by encouraging resilient and varied ecosystems. Preventing forest degradation can be accomplished through sustainable measures including protecting young trees and selective logging. Using harvesting techniques that do not harm the forest environment is essential to the long-term viability of silvicultural operations.			
Operation and maintenance	Forest silviculture requires careful management to adapt to the climate along with the challenging terrain of the area. To maintain robust development, key actions include planting trees, conserving soil, and controlling pests and diseases. Selective logging allows for sustainable harvesting, and community involvement ensures local participation in forest management, fostering both ecological sustainability and better livelihoods.			
Cost	Medium			
Benefits				
	Type	Low	Medium	High
	Canopy Cover Increase			
	Carbon Sequestration			
	Crop Production Increase			
	Crop Failure Risk Reduction			
	Ensuring Food Security			
	Fruit Production Increase			
	Forestry Goods Production Increase			
	Livestock Production Increase			
	Fisheries Production Increase			
	Alternative Livelihood Generation			
	Increase of Income			
	Soil Erosion Control			
	Improved Soil Health			
	Soil Fertility Increase			
	Sustainable Water Supply			
	Water Quality Improvement			
	Water Purification			
	River Bank Erosion Control			
	Surface Runoff Reduction			
	Groundwater Recharge			
	Pest and Disease Management			
	Reduction of Pesticides and Fertilizer Use			
	Ecosystem and Biodiversity Increase			
	Better Communications			
	Better Education			

	Better WASH facilities			
Implementation Challenges				
	Type	Low	Medium	High
	Land unavailable/high land area required			
	Supply chain disconnection			
	Land right related problems			
	High human investment capital need			
	Insufficient infrastructures			
	Less demand of agroforestry product			
	Environmentally unsustainable			
	Socially unacceptable			
	Less beneficial			
Environmental performance	• In addition to preventing soil erosion on steep slopes by using erosion control techniques like terracing, mulching, and contour bunds, forest silviculture operations also aid in monitoring the water cycle by protecting catchment areas and retaining water, which prevent flooding and soil degradation. By maintaining soil fertility, good forest management techniques increase land production and lessen the need for chemical inputs. • Forest silviculture assists conserve carbon by preserving healthy tree growth, which lessens the effects of climate change. protecting biodiversity and providing habitat for wildlife by fostering native species and preserving mixed-species forests. • Integrated pest management lessens the negative effects of chemical treatments on the environment while preserving healthy forests. The sustainable use of forest resources is ensured via controlled harvesting cycles and selective logging, which permits natural regrowth.			
Remarks	This measure will aid landscape restoration in CHT of Bangladesh along with creating alternative livelihood opportunities.			